

# Lee Wen Yeong

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## EDUCATION

Bachelor of Science - National University Of Singapore | Business Analytics (School Of Computing)

- Machine Learning Analyst & Co-Training Head, NUS Fintech Society

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## INTERNSHIP EXPERIENCE

### Research Assistant, Karlsruhe Institute Of Technology

April 2026 -

- 4D Gaussian FFN for Dynamic Driving Environments: Architecting a novel, memory-efficient streaming network for real-time 3D reconstruction in complex driving scenarios; optimizing for high-frequency temporal consistency and low-latency inference. Manuscript in preparation for submission (Target: End of 2026).

### Student Ambassador, Nvidia

Sep 2025 -

- Architected an end-to-end computational biology platform leveraging NVIDIA NIMs for high-throughput protein structure prediction and drug discovery workflows.
- Optimizing AlphaFold3 Inference: Developed custom CUDA kernels to eliminate memory-bound bottlenecks, achieving significant gains in GPU throughput; research currently under preparation for publication.

### Machine Learning Engineer Intern, TikTok (Search Algorithms Team)

June 2025 - February 2026

- Developing a novel multi-modal ranking model strategy in C++ for user search, integrating it directly into the search engine's retrieval pipeline to unify user wiki, account, and video data into a unified "big card" result, targeting a 5% improvement in search relevance and user engagement.
- Developed in house agent to compare difference in strategies, leading to a reduction in man hours using MCP tool and langchain. Implemented kv caching system to reduce latency and increase scalability
- Developed an end-to-end recall strategy utilizing knowledge distillation from a large-scale Qwen-3.5 model into a 0.7B parameter version, enabling low-latency, scalable word suggestions for TikTok Tako.

### Data Science & AI Intern, Ministry Of Manpower

June 2024 - August 2024

- Elevated investigator efficiency by over 60% through the optimization of an internal LLM used to aid in work-related incidents via techniques like Chain of Thought and Supervised Fine Tuning

### Computer Vision & Video Analytics Intern, DSTA

February 2023 - July 2023

- Spearheaded the development and successful deployment of a novel in-house computer vision algorithm for real-time identity tracking at NDP 2023 using libraries like PyTorch & OpenCV, significantly enhancing operational intelligence for thousands of attendees.
- Researched and presented cutting-edge data engineering solutions for a joint international challenge (US DoD, DSTA, ST Engineering), showcasing deep expertise in large-scale data processing and analytics at the Singapore Defence Tech Summit 2023.

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## PROJECTS

### 1. Flash-Attention in metal cpp:

Architected a high-performance FlashAttention forward pass in metal-cpp, achieving up to 3.8x speedup over MLX and 82x over PyTorch for 2048-token sequences. Optimized memory bandwidth and compute utilization via custom MSL (Metal Shading Language) kernels.

### 2. Person Re-Identification Project

Constructed a multithreaded person re-identification system leveraging YOLO-NAS and a CTL (Centroid Triplet Loss) model with ResNet-50, achieving a remarkable 98% accuracy for robust identity tracking.

**Research Paper Reference:** [arXiv Paper](#)

### 3. OBrainRot: (116 stars on Github!)

Orchestrated the development and deployment of a full-stack AI application automating "brain rot" video generation from Reddit, integrating Llama 3.3 & VADER for sentiment analysis, and a robust audio-video pipeline (Coqui xTTSv2, Wav2Vec2, FFmpeg, custom overlays) all built with React, Flask, Python, and Docker.

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## CERTIFICATIONS

- Google Advanced Data Analytics Certificate
- MySQL Data Analytics & Business Intelligence By Udemy
- AI4I Level 2

**TECH STACK:**

- Programming Languages: Python, Java, Javascript, C#, C++
- Machine Learning: Tensorflow , Pytorch, Scikit-learn, CuPy, OpenCV, CUDA
- Web Development: React, Flask, Node.js, Django
- Databases: MySQL, MongoDB
- Others: HTML, CSS, Git, Docker, Kubernetes